

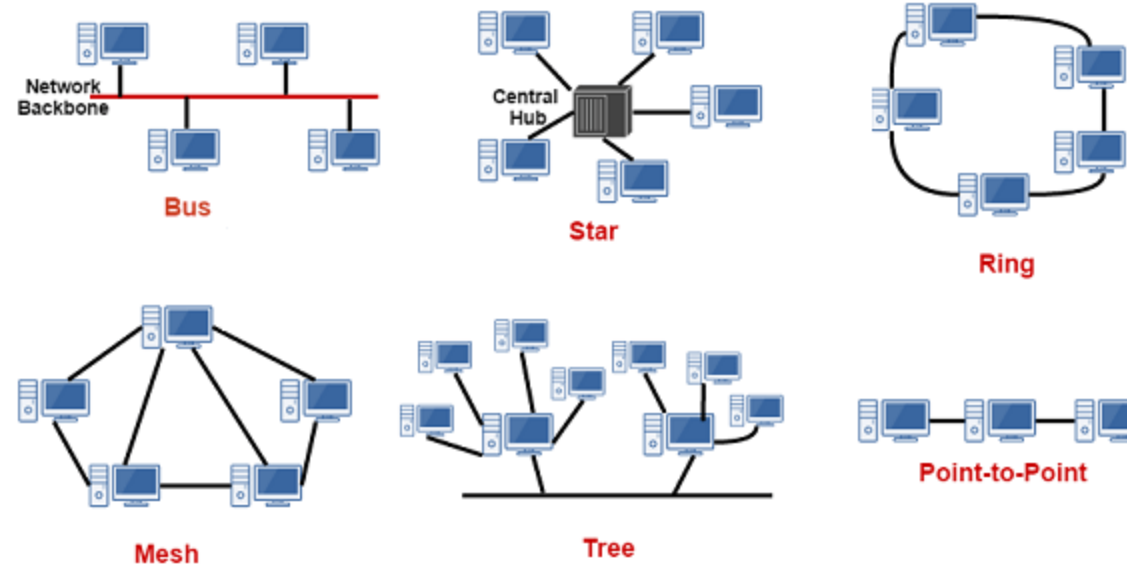
Web Services

What is a Network?

- A computer network consists of ***two or more computers that are linked in order to share resources*** (such as printers and CDs), ***exchange files***, or ***allow electronic communications***.
- The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

What is a Network Topology?

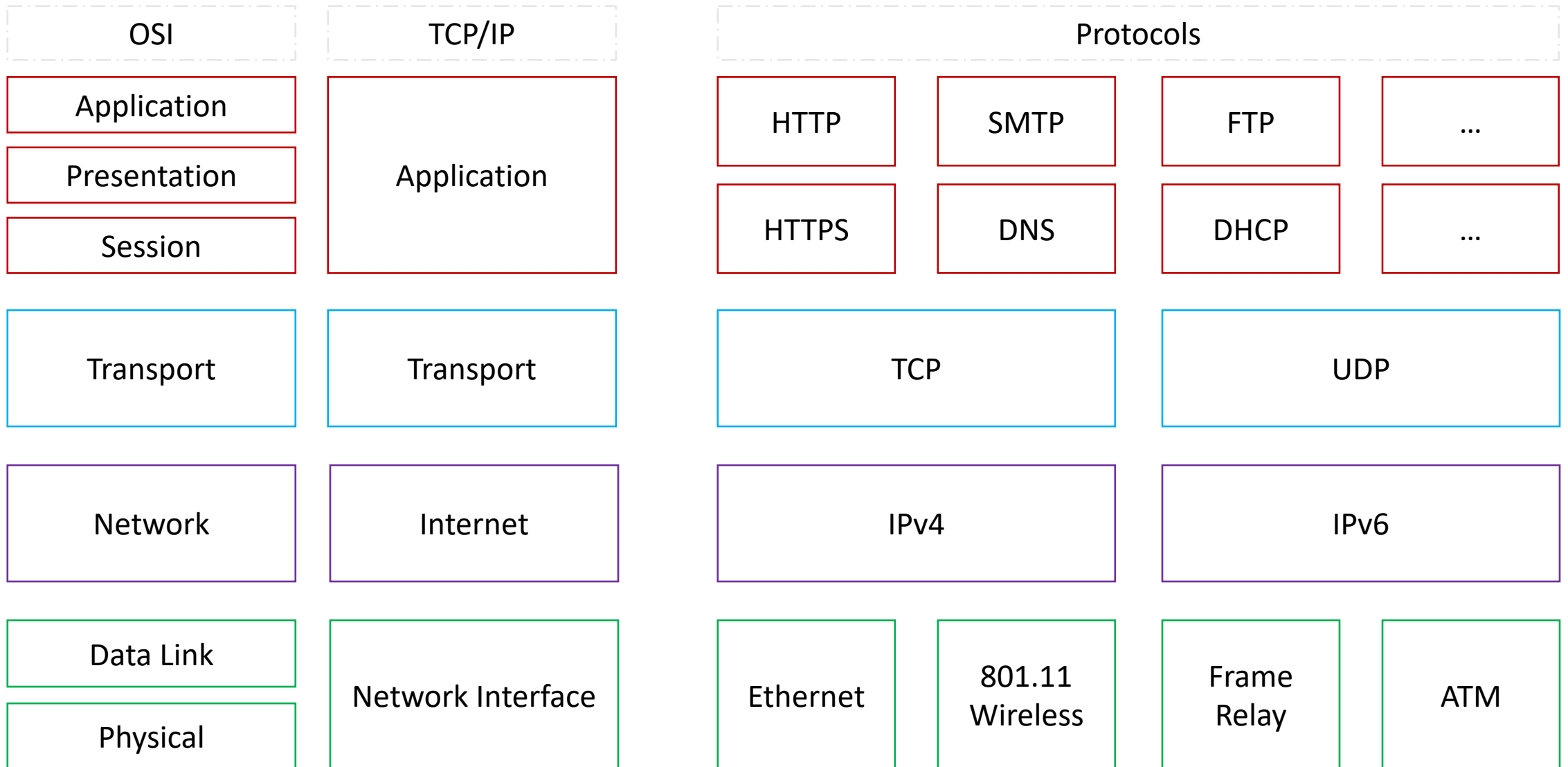
- Network topology is the arrangement of the elements (links, nodes, etc.) of a communication network.



What is a Network Protocol?

- A network protocol is an established ***set of rules that determine how data is transmitted*** between different devices in the same network.
- Essentially, it allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure or design.

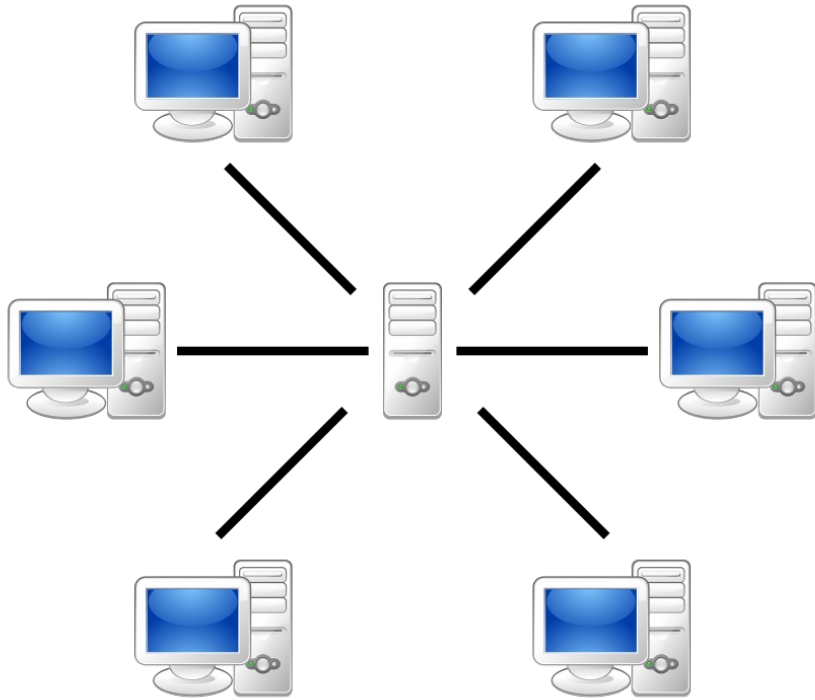
Network Protocols



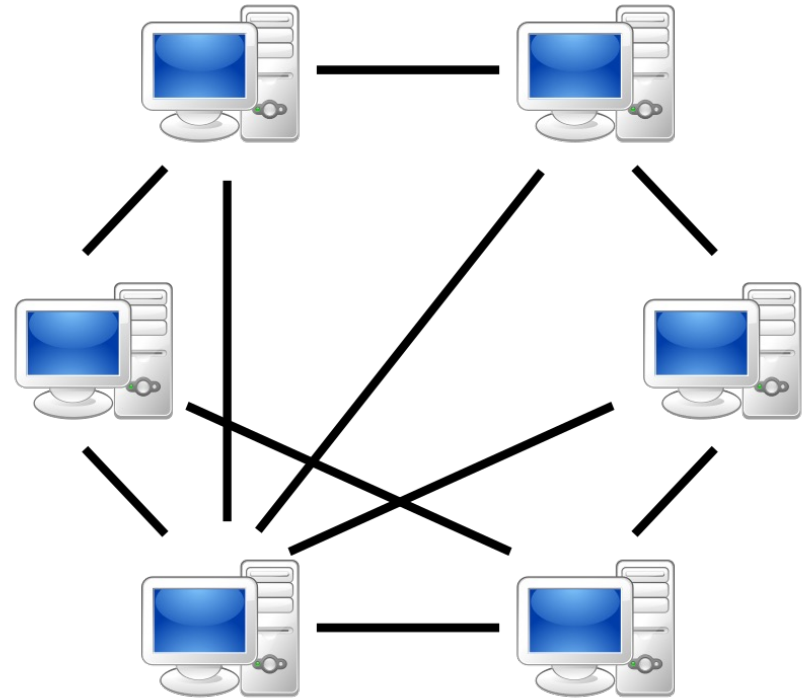
TCP/IP - Application Layer Protocols

- The application layer contains the ***communications protocols*** used in ***process-to-process*** communications across a computer network.
- The application layer ***only standardizes communication*** and depends upon the underlying transport layer protocols to establish host-to-host data transfer channels and manage the data exchange in a client-server or peer-to-peer networking model.
- The application layer ***does not describe specific rules or data formats*** that applications must consider when communicating.

Network Types



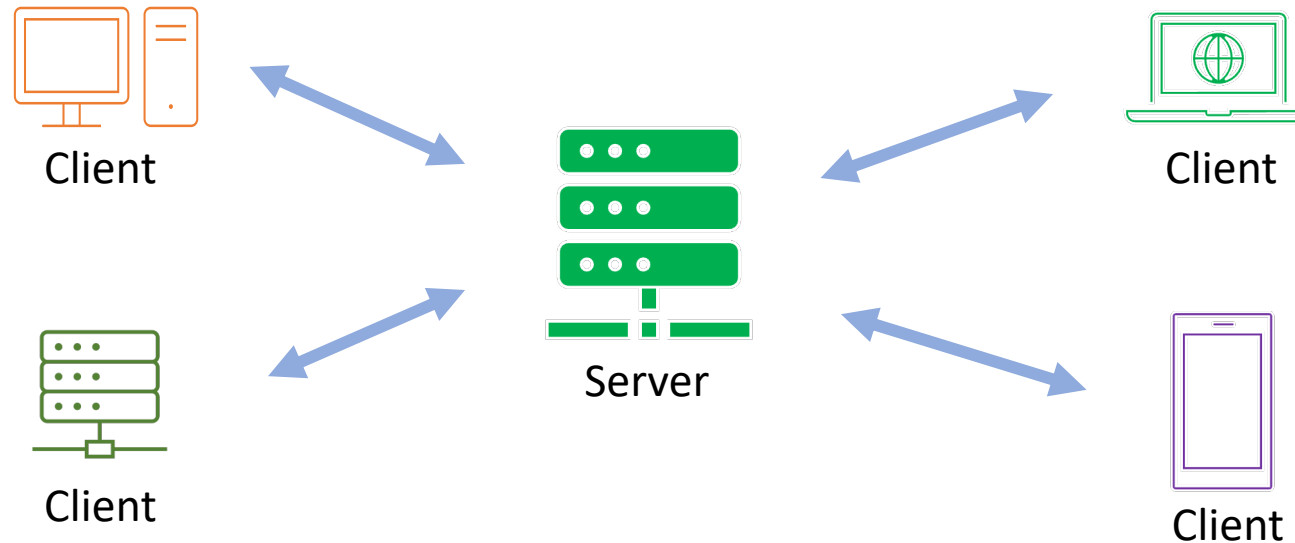
Client Server



Peer to Peer

Client Server

- Client Server Architecture is a computing model in which the server hosts, delivers and manages most of the resources and services to be consumed by the client.
- ***This type of architecture has one or more client computers connected to a central server over a network or internet connection.***



What is the Internet?

- The Internet is the global system of ***interconnected computer networks*** that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices.
- ***It is a network of networks.***



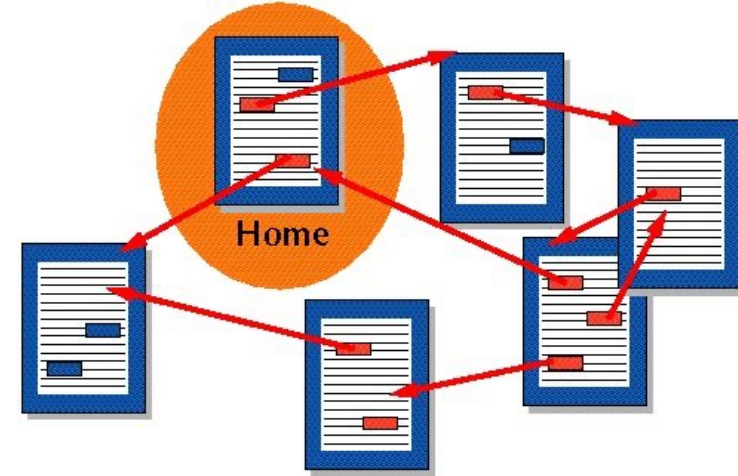
What is the Web?

- The World Wide Web (WWW), commonly known as the Web, is an information system.
- Where *resources (documents, images, videos, ...)* are *identified by Uniform Resource Locators **URLs***, which are *accessible over the Internet*.
- ***The World Wide Web** is built on top of the Internet, which pre-dated the Web by over two decades.*



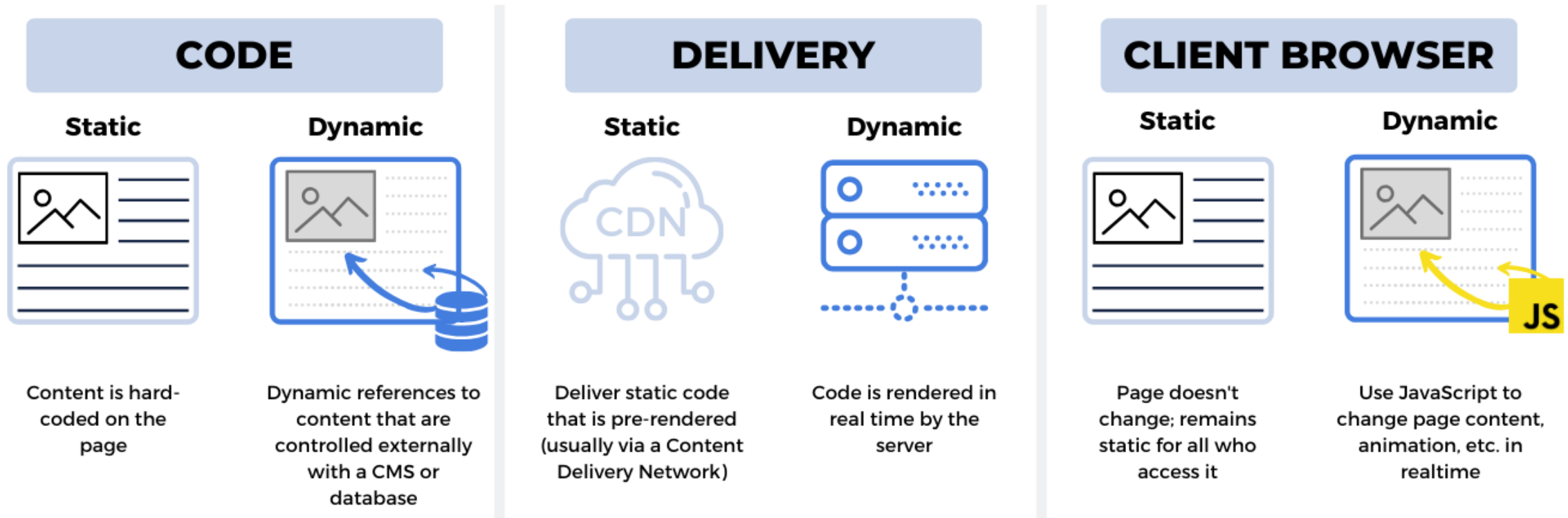
Web Page & Web Site

- A **Web Page** is a *hypertext document* provided by a website and displayed to a user in a web browser.
- A **Web Site** is a collection of web pages and related content that is identified by a common domain name.
- All publicly accessible websites collectively constitute the World Wide Web.

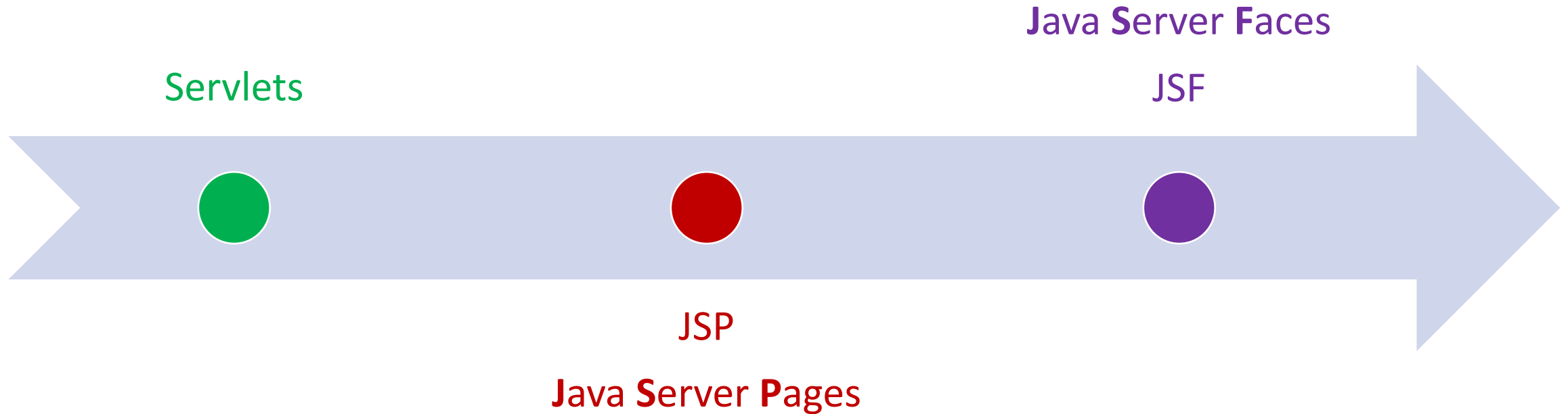


Static vs Dynamic Web Site

- Static websites are ones that are ***fixed and display the same content*** for every user, usually written exclusively in HTML.
- A dynamic website, ***displays different content and provide user interaction***, by making use of advanced programming and databases in addition to HTML.



Dynamic Web Site Development Technologies in Java



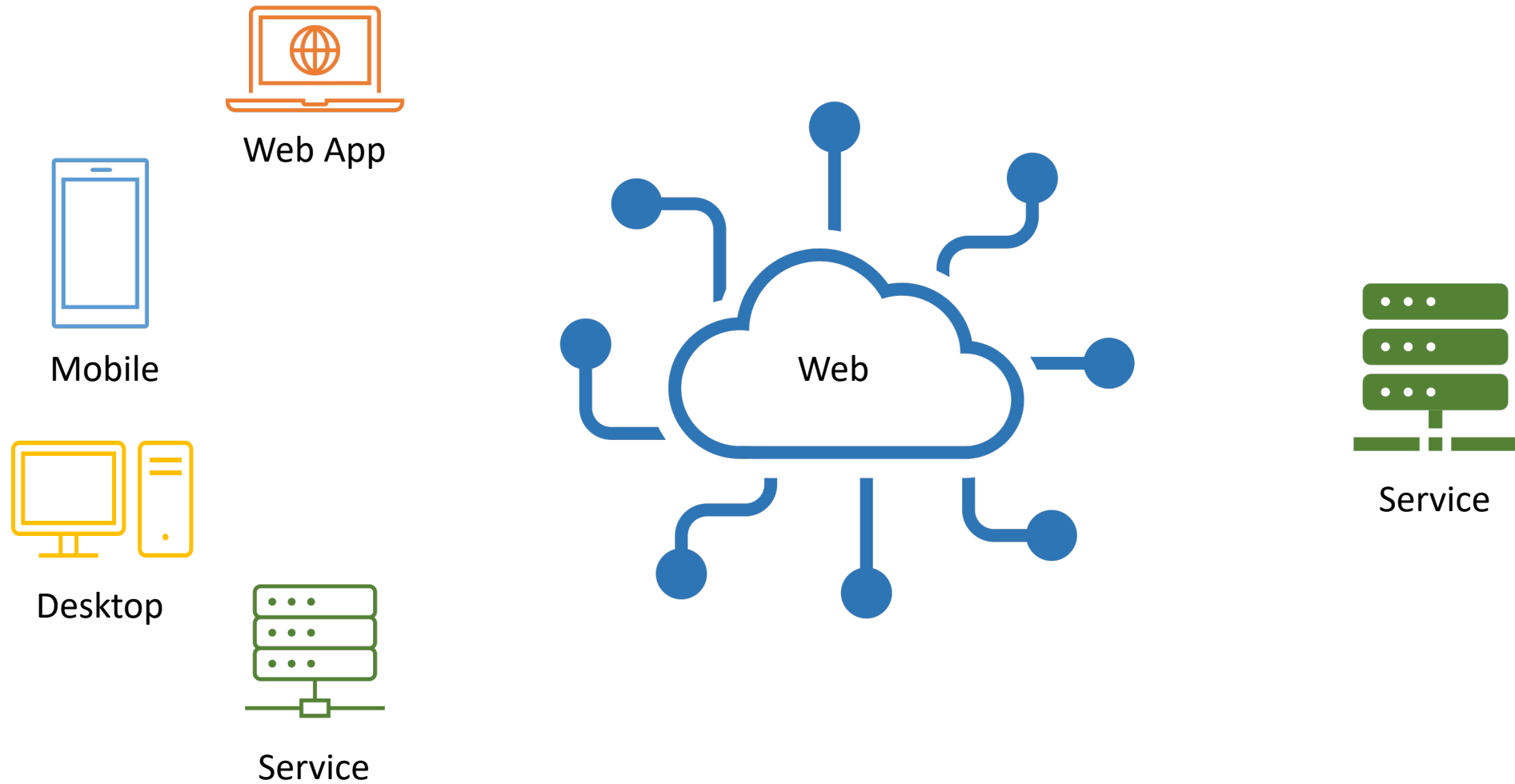
What is a Service ?

It's a Functionality
provided by a Software, to other Software

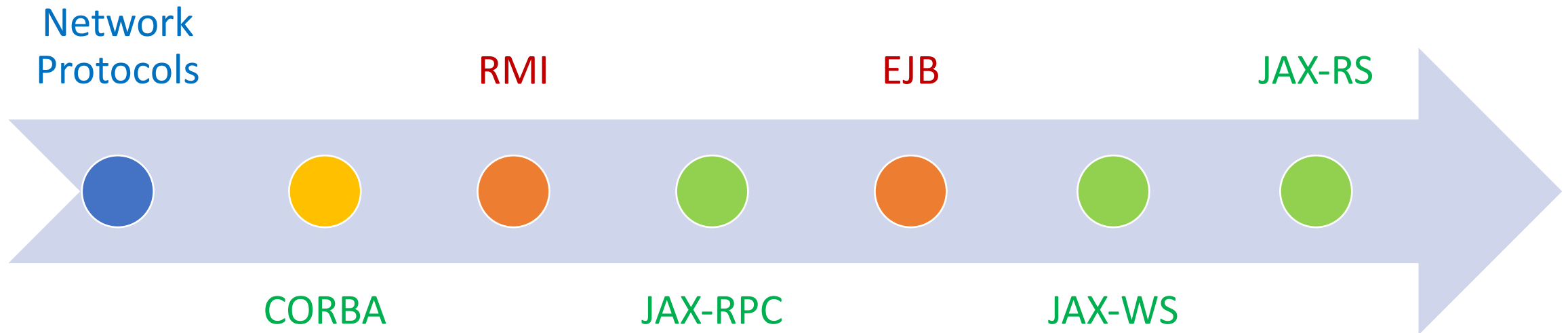
What is a Web Service ?

It's a Service
that's accessible over the Web

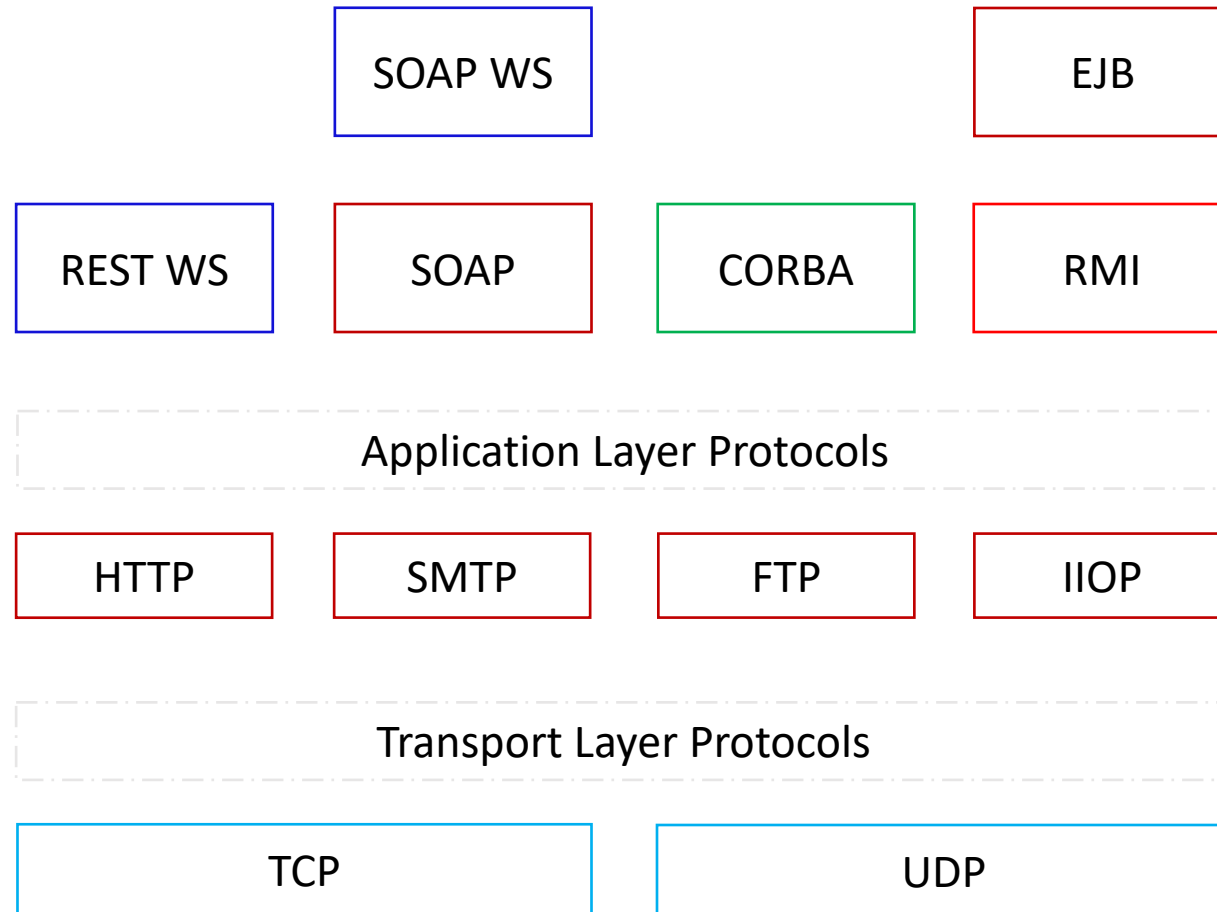
Web Services



Web Services Development Technologies in Java



Web Services & Network Protocols

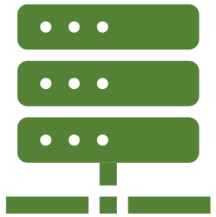


Web Applications **vs** Web Services



Web Applications

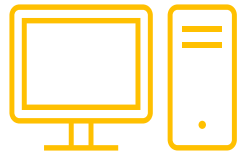
Meant For Human Consumptions



Web Services

Meant For Machine Consumptions

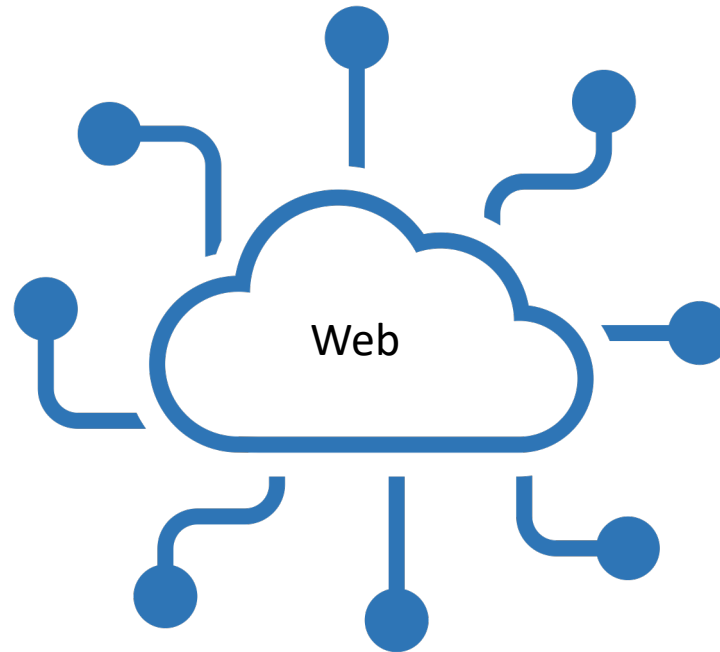
SOAP Web Services



Desktop

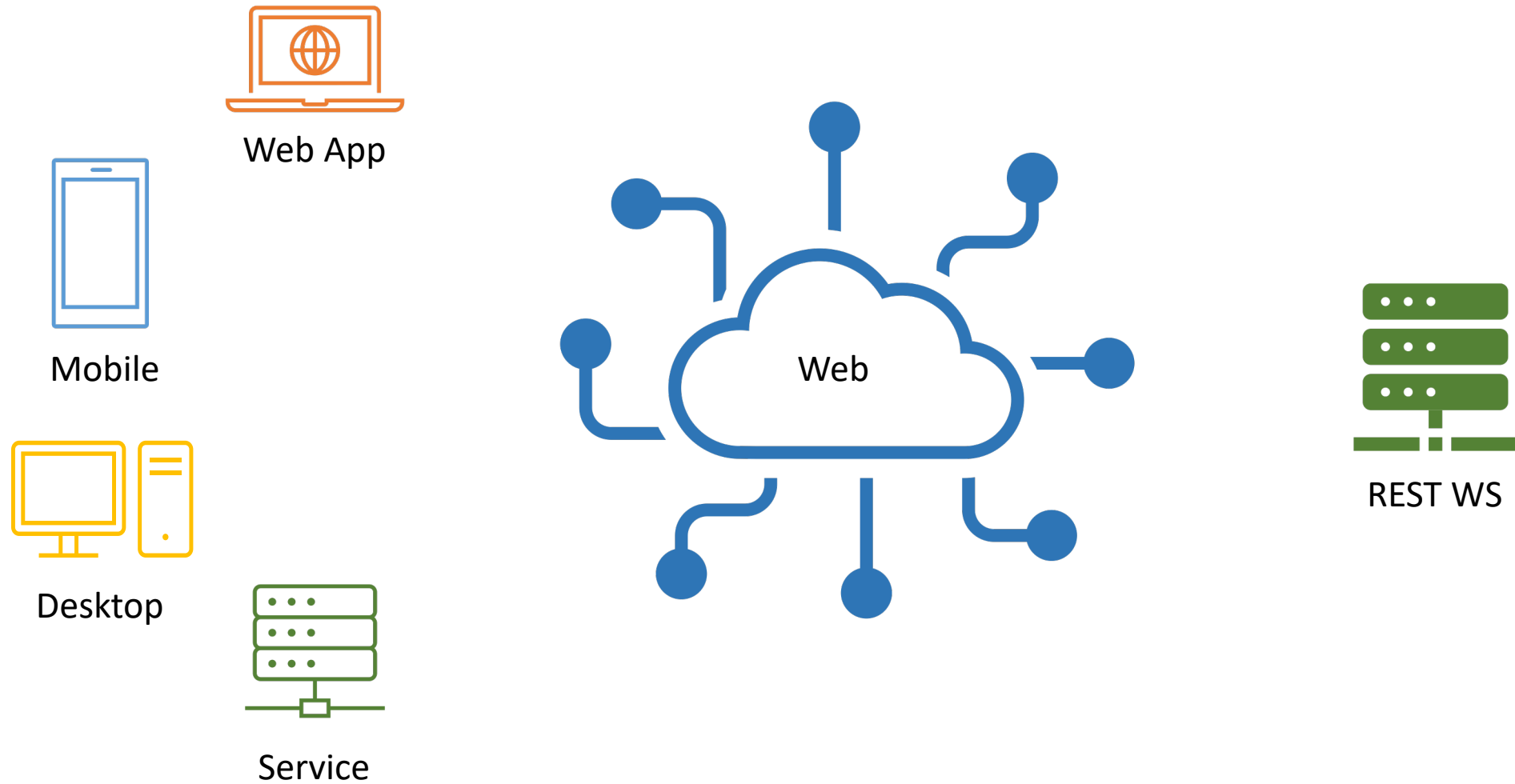


Service

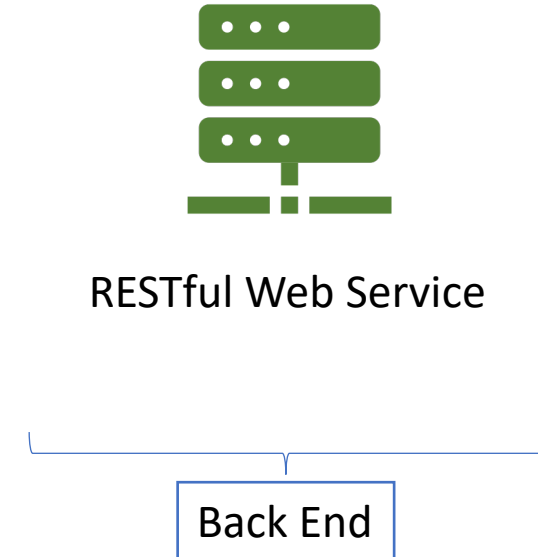
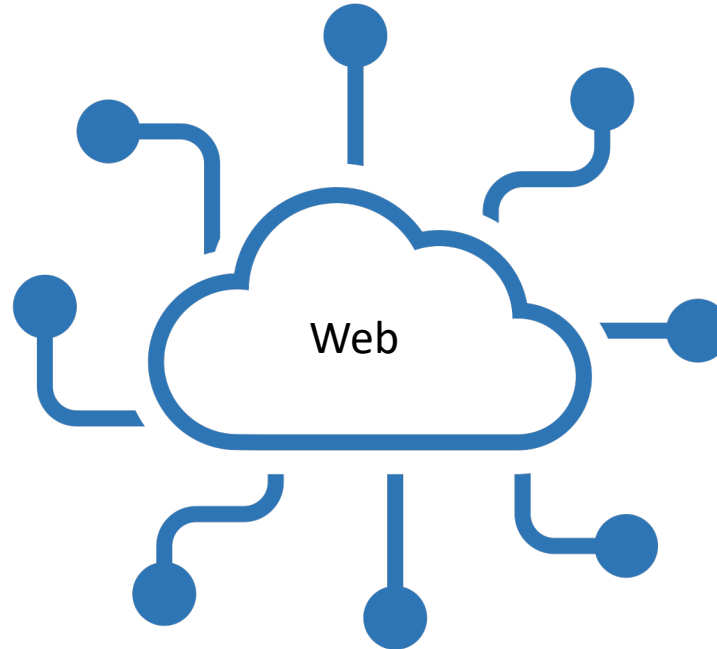
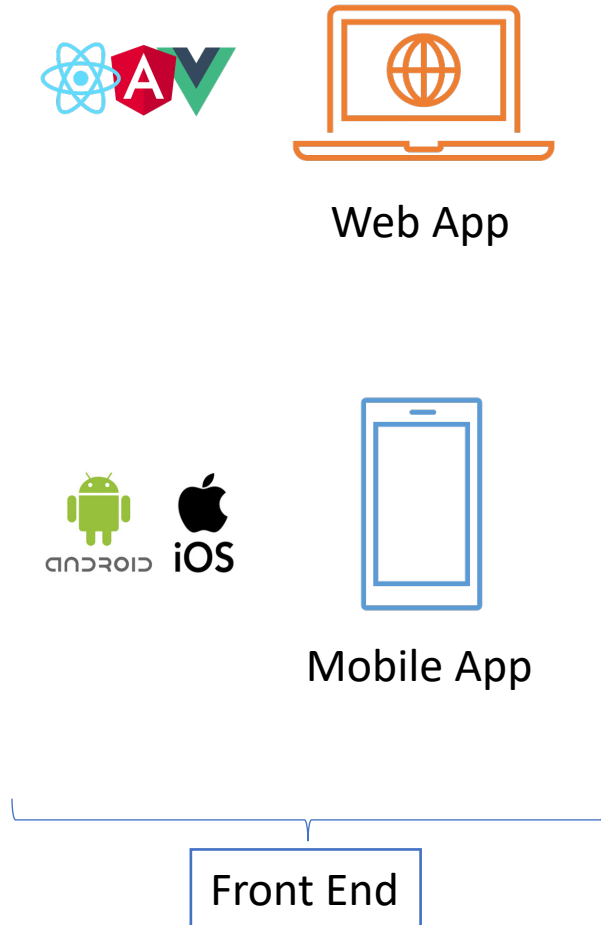


SOAP WS

REST Web Services



Modern Web & Mobile Applications

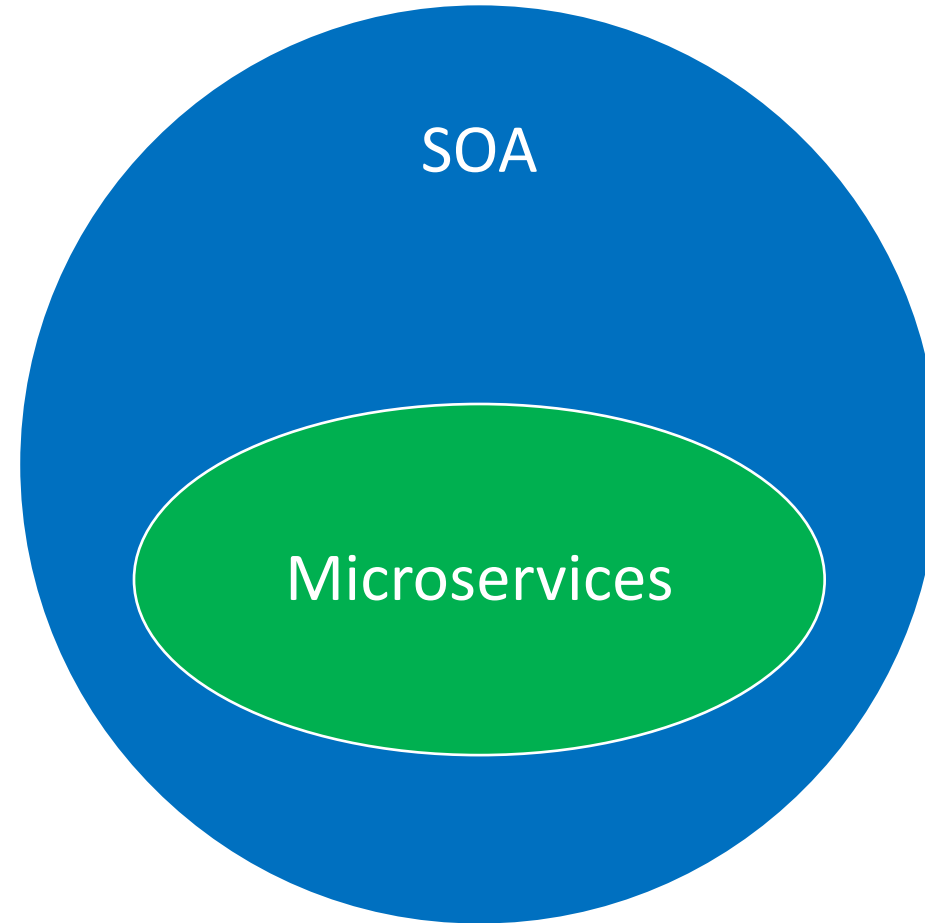


Service Oriented Architecture



A Single Application Developed as a Set of Communicating Web Services

Microservice Architecture



Web Services Design

